



Software Testing and Quality Assurance
Second Semester 2025/2026
Homework Assignment # 2
Total Mark (20 points)

1. Go to the source code and Download it:
<https://anonymous.4open.science/r/Junit5-HW-50D4/>
2. Import the project
3. You Need to do the following tasks
 1. Write test classes for each class in the **main.najah.code**: (4 points)
 - Calculator (**TestCalculator**)
 - Product (**TestProduct**)
 - UserService (**TestUserService**)
 - RecipeBook (**TestRecipeBook**)
 2. Each test class should:
 - Test both valid and invalid input (at least 1 test method for each method, 2 points)
 - Use `@DisplayName` to make tests descriptive (1 point)
 - Include at least one parameterized test (2 points)
 - Include a timeout test (1 point)
 - Try to use multiple assertions (at least 2 assertions 2 points I should at least see `assertEqual`, `assertThrows`)
 3. Create a test **suite** using `@Suite` that runs all test classes together. (2 points)
 - I should see (`@Suite`, `@IncludePackages()` or `@SelectClasses()`)
 4. For CalculatorTest, demonstrate ordered test execution using `@Order`.
 5. **Lifecycle Hooks**
 - Use `@BeforeAll`, `@AfterAll`, `@BeforeEach`, and `@AfterEach` in at least one test class and print simple debug messages (e.g., "setup complete"). (2 points)
 6. Include one intentionally failing test and **disable** it with `@Disabled`, with a note on how to fix it. (1 point)
 7. Use `EclEmma` (or equivalent) to measure your **test coverage**. Try to get above ~85% for each class.
 8. Enable parallel execution in JUnit 5 and apply `@Execution(ExecutionMode.CONCURRENT)` in one test class. (2 points)